



# Solar and Lunar Eclipses



## Learning Outcomes

### The learner:

- explains the broad reasons why eclipses happen.
- differentiates between solar and lunar eclipses.
- learns about precautions to be taken while viewing an eclipse.



## Let's Take-off

Creativity



What do you wonder about when you see these pictures of nature?



Share the thoughts of wonder that came into your mind with your friend in class. One thought could be—why does the sky become orange and yellow when the sun rises or sets?

Human beings have always wondered about nature and tried to find explanations for the things that they see. One thing that people tried to find an explanation for was a **phenomenon** where the light of the sun or the moon was partially or fully hidden. People got frightened when they saw a sun like the one in the picture here. They made up all kinds of stories. However, none of these was scientific.



The sun, partially hidden in an eclipse

## ECLIPSES

When the light of one heavenly body is partially or completely hidden from us by another heavenly body which is in the way, it is called an **eclipse**.

**Astronomers** and scientists have now explained how eclipses happen. They observed, studied and understood eclipses and have given scientific reasons, some of which are listed below.

### Reasons for Eclipses

Eclipses happen for the following reasons:

- the orbit of the moon around the earth
- the orbit of the earth around the sun
- the shadow that is formed when sunlight falls on an **opaque** object (opaque=something through which light cannot pass)



#### Discuss

Communication



Why do you think that eclipses frightened people?

C-6.1



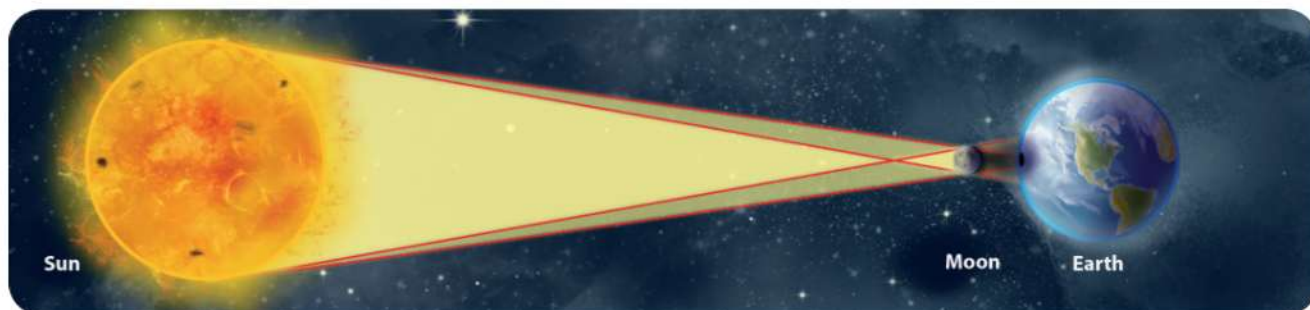
#### Knowledge of India

One of the first people who explained how eclipses happen was Aryabhata, an Indian mathematician and astronomer who lived in the 5th and 6th centuries CE.

## SOLAR ECLIPSE

'Sol' means the sun. A **solar eclipse** happens when the sun, moon and the earth are in a line with the moon between the earth and the sun. The moon casts a shadow on the earth as it hides the light of the sun from the earth.

When all the three heavenly bodies are in a perfect straight line, with the moon between the sun and the earth, we have a total solar eclipse. The sky becomes very dark, as though it is night, and birds stop singing. A total solar eclipse can only be seen from a small part of the earth.



A solar eclipse—Locate the place on the earth where the moon is casting a shadow.

We see a partial solar eclipse when the three heavenly bodies are not in a complete straight line. Solar eclipses last for only a few minutes. They are exciting, but we should never look at an eclipse of the sun directly, even with dark glasses.



A total solar eclipse

Special equipment called **solar viewing glasses** or eclipse glasses or a safe solar viewer must be used to view either a partial or a total solar eclipse. These glasses or viewers are made to certain safe standards. They are not the regular dark sunglasses that one uses during summer. One must check the solar glasses for scratches or damages before use so that our eyes remain safe. Cameras, binoculars or telescopes should not be used as the strong rays of the eclipse can burn the filter and cause serious eye damage.



### Did You Know?

On average, total solar eclipses happen only once every 18 months and the longest time that a solar eclipse can last is around 7 minutes 30 seconds.

## LUNAR ECLIPSE

'Luna' means the moon, so **lunar eclipse** is the eclipse of the moon. This happens when the moon goes behind the earth while moving round it. The earth then lies between the sun and the moon. The shadow of the earth falls upon the moon, eclipsing it and preventing the sun's light from falling on the moon.

The diagram here will show you how it happens.



Lunar eclipse—how can you say that the moon has been eclipsed?

In a total lunar eclipse, the moon and the sun are on either side of the earth in a perfect straight line. The shadow of the earth falls on the moon. The moon does not appear dark but looks slightly reddish during a total lunar eclipse.

In a partial lunar eclipse, only a part of the moon moves into the earth's shadow. The earth's shadow appears very dark.

Unlike a solar eclipse, a lunar eclipse usually lasts longer, sometimes a few hours. It is safe to look at a lunar eclipse, so we do not need special equipment to look at it.



A total lunar eclipse



A partial lunar eclipse



### HOTS

Critical Thinking

21st  
CENTURY  
SKILLS

At what time of the day can you see a lunar eclipse? **C-6.1**

Eclipses are beautiful natural phenomena. Science has made it possible for us to understand these natural phenomena. Unfortunately, sometimes, because of cloudy weather or because of pollution, we are not able to see an eclipse.



## Think and Answer

Critical Thinking

21<sup>st</sup>  
CENTURY  
SKILLS

Why is it safe to look at a lunar eclipse but unsafe to look at a solar eclipse?

C-6.1



## A Better World

Sustainable Development Goals (SDGs)

### Goal 12: Responsible Consumption and Production

The solar or the lunar eclipse can sometimes not be seen because of pollution of the atmosphere. What can we do to reduce air pollution and keep our air and skies clean and clear?

C-6.1



## Values and Life Skills

- Why should we not believe olden stories of why eclipses happen?
- How can you help others to see the beauty of an eclipse and also remain safe?

## REMEMBER

### Solar and Lunar Eclipses

- The orbits of the moon and the earth and their shadows are the reason for eclipses.
- Solar eclipses happen when the moon moves between the sun and the earth.
- It is dangerous to look at a solar eclipse without special equipment.
- Lunar eclipses happen when the earth moves between the sun and the moon.
- Eclipses may be complete or partial.



## NEW WORDS

**Phenomena**

a significant event in nature; phenomena is the plural of phenomenon

**Astronomer**

a scientist who studies the sun, moon, stars, planets, etc.

**Opaque**

not clear enough to allow light through, the opposite of transparent



## EXERCISES

Remembering and Understanding

### A. Choose the correct answer.

1. In ancient times, most people had **logical/scientific/unscientific** explanations for why an eclipse happened.
2. A solar eclipse is when the **sun/moon/earth** is completely or partially hidden.
3. A lunar eclipse is when the **earth/sun/moon** is completely or partially hidden.
4. A lunar eclipse lasts for a few **minutes/hours/days**.
5. Read the statements below and choose the correct option.

**Statement (A):** It is unsafe to look a solar eclipse.

**Statement (B):** Solar eclipse is the eclipse of the sun.

- (a) Both A and B are true, and B is the correct explanation of A.
- (b) Both A and B are true, but B is not the correct explanation of A.
- (c) A is true but B is false.
- (d) A is false but B is true.

### B. Write true or false. Correct the false statements in your notebook.

1. 'Sol' means the moon and 'luna' means the sun.
2. During a total solar or lunar eclipse, the three heavenly bodies—the sun, the moon and the earth—are in a straight line.
3. It is not safe to look at a solar eclipse even with dark glasses.
4. Solar and lunar eclipses both last for the same amount of time.
5. We now know how and why eclipses happen, so we should not be frightened of them.

### C. Fill in the blanks.

1. An ..... happens when shadows fall on the earth or the moon.
2. When light falls on an ..... object, the shadow falls on the opposite side.
3. In a ..... eclipse, the moon is between the sun and the earth.
4. A ..... eclipse is unsafe to see without protection.
5. We must only believe ..... explanations about eclipses.

### D. Match the following.

| A              | B                                                  |
|----------------|----------------------------------------------------|
| 1. Sol         | (a) Being hidden                                   |
| 2. Luna        | (b) The moon                                       |
| 3. Eclipse     | (c) Scientists who study the sun, moon, stars etc. |
| 4. Astronomers | (d) The sun                                        |

### E. Answer the following questions.

1. How do we know the real reason why eclipses happen?
2. Give the three reasons why eclipses happen.
3. Why should we not look at the sun directly during a solar eclipse?
4. In two sentences describe what happens during a partial lunar eclipse.
5. Sometimes we are not able to see solar eclipses that occur. Why is this?



## DO AND LEARN



### A. Picture Study

Describe the pictures. What are they showing?

C-2.1 Critical Thinking



TP

### B. On Your Own

- Find out about the special glasses required to view an eclipse. Get one of them for your home.
- Find out when the last total solar eclipse happened and when the next one is going to be.

C-6.1 Research-based

### C. In a Group

- Do this experiment

*You will need:*

- |                                       |                                 |
|---------------------------------------|---------------------------------|
| (a) a small round grape (as the moon) | (e) a sweet lime (as the earth) |
| (b) a toothpick                       | (f) a little modelling clay     |
| (c) a torch (as the sun)              | (g) a big ruler                 |
| (d) a table to do the experiment on   |                                 |

*What to do:*

Do this experiment in a dark room.

- Stick the toothpick on the modelling clay so that it is about 4.5–5 cm high. Put the grape on top of it and place it upright on the table.
  - Put the sweet lime on the table, about 8 cm behind the grape.
  - From about 30 cm away, shine the torch on the grape. Make sure that the sweet lime, the grape and the torch are in a straight line. What do you see? Talk about this in class.
- Go out into the sunlight in the morning or the evening. Ask a friend to mark the length of your shadow. Do the same at noon. When was your shadow shorter? Why is this so?

C-6.1 Experiential learning SI Science

C-6.1 Collaboration

- ### D. Pair and Share:
- Discuss with your partner in class why it is important to understand the scientific reason for a natural phenomenon.

C-6.1 Communication



**Weblinks** For more information, go to:

- Solar eclipse** <https://www.youtube.com/watch?v=E6OtLfszaVI> (Accessed on 31 July 2025)
- Lunar eclipse** <https://www.youtube.com/watch?v=LH-aSizsx1o> (Accessed on 31 July 2025)



## Self-review

### What I Learnt

| Colour the smileys to show how well you have understood the concepts. | Completely | Mostly | Need help |
|-----------------------------------------------------------------------|------------|--------|-----------|
| I can explain the broad reason why eclipses happen.                   |            |        |           |
| I can differentiate between solar and lunar eclipses.                 |            |        |           |
| I can learn about precautions to be taken while viewing an eclipse.   |            |        |           |